

What Don't We Know: Identifying Specific Gaps in GenAI Literacy

Proposed Features for Learning Support Tools in AI Systems

Introduction

Purpose

- Generative AI (GenAI) is proliferating throughout professional and academic contexts.
- GenAI can be used to great effect BUT used carelessly can be to great detriment

What do you think the average person knows about GenAI?

Background

- Kopkow et. al. (2025) found in the context of Austrian small and medium sized enterprises (SMEs) concerning trends in GenAI literacy
 - Mean GenAI literacy score of 7.6/12
 - An average of 1.7/3 for 2 of the 4 dimensions of GenAI literacy
 - 83.7% of participants report daily or weekly GenAI use

Workers are using GenAI without a full grasp of what it can do and its limitations

Research Questions & Hypothese

- **RQ1**– Which specific concepts are most frequently misunderstood, with regard to participants' GenAI literacy?
 - **H1**: We will find significant differences between the dimensions of the GLAT.
- **RQ2**– What concerns do participants currently have regarding GenAI, and are they congruent with the GenAI literacy (i.e. do the concerns match the level of skill with GenAI shown by the GLAT)?
 - **H2**: We will find that most concerns are subjectively not reflective of participants GenAI literacy.

Methodology

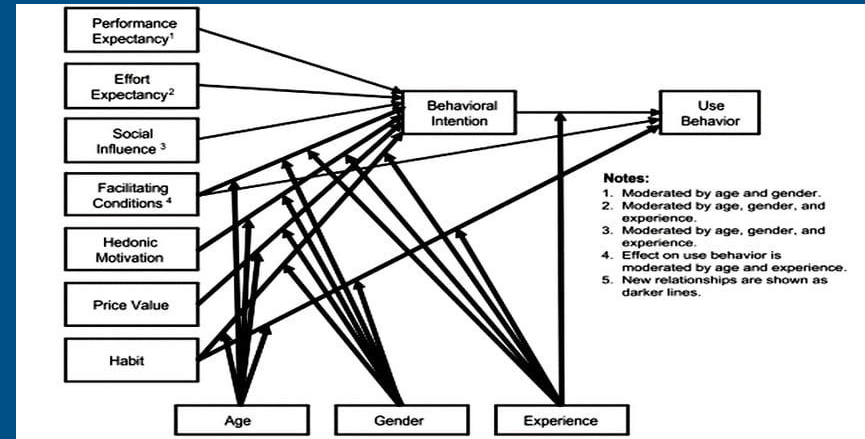
Dataset 1

- **The Generative AI Literacy Assessment Test (GLAT)**
 - $N = 50$
 - 12 single choice questions over 4 dimensions of GenAI Literacy
 - **Know & Understand:** GenAI structure and architecture, as well as the general capabilities of a GenAI model
 - **Use & Apply:** What tasks GenAI models can do and how they do them
 - **Evaluate & Create:** How to supervise and evaluate the output of a GenAI model in order to create accurate, high quality content
 - **Ethics:** GenAI limitations and ethical issues that may arise from use of GenAI models

Dataset 2

- **Online Survey**
 - N = 34
- **2 Open-ended questions**
 - “How do you currently use applications that incorporate GenAI into your day-to-day work?”
 - “What concerns or worries do you have in your day-to-day work regarding the use of applications that incorporate GenAI?”

- **Unified Theory of Acceptance and Use of Technology 2 (UTAUT2)**
 - 6 constructs which predict behavioral intention to use a technology



The Analysis

- GLAT
 - Descriptive statistics (Whole Scale)
 - Descriptive statistics (4 Dimensions)
 - Friedman rank sum + Wilcoxon rank sum post-hoc (Alternative to within subjects ANOVA)
 - Item level frequencies
- UTAUT2
 - Descriptive statistics (6 Constructs)
- Open-Ended Questions
 - Inductive content analysis

Results

Whole Scale

Mean	Median	Standard Deviation	Minimum	Maximum
6.84	7	2.486	1	11

By Dimension

Dimension	Mean	Median	Standard Deviation
<i>Know & Understand</i>	2.14	2	0.857
<i>Use & Apply</i>	1.38	1	0.967
<i>Evaluate & Create</i>	1.64	2	1.045
<i>Ethics</i>	1.68	2	1.096

GLAT: Dimensions

Friedman Rank Sum

χ^2	df	p (<.001)
18.32	3	<0.001***

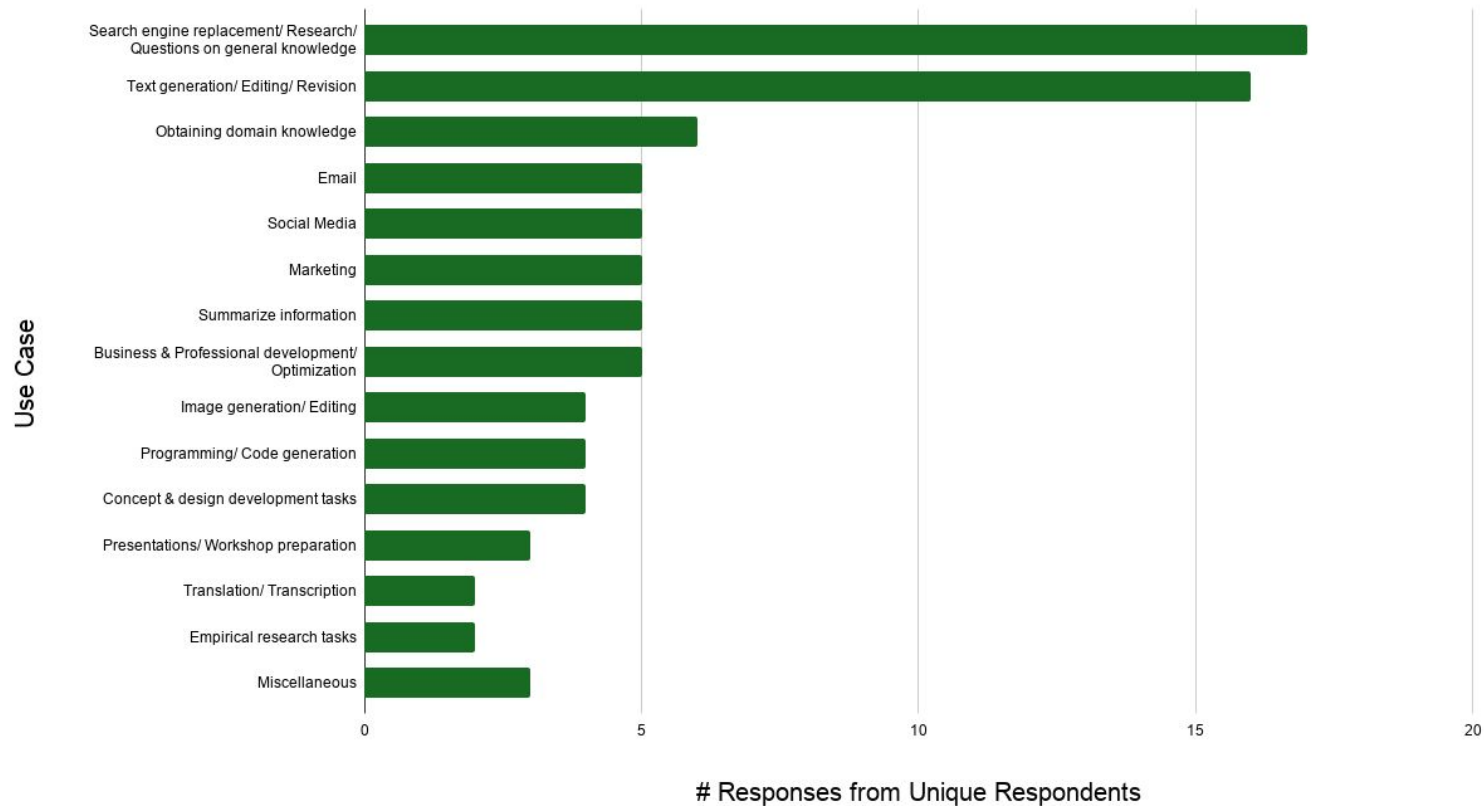
Wilcoxon Rank Sum: p values

Dimensions	<i>Know & Understand</i>	<i>Use & Apply</i>	<i>Evaluate & Create</i>
<i>Know & Understand</i>	–	–	–
<i>Use & Apply</i>	0.001***	–	–
<i>Evaluate & Create</i>	0.098*	1.000	–
<i>Ethics</i>	0.224	0.924	1.000

Construct	Mean	Median	Standard Deviation	Minimum	Maximum
<i>Performance Expectancy</i>	5.245	5.333	0.712	3	6
<i>Effort Expectancy</i>	4.066	4.250	1.173	1.25	6
<i>Social Influence</i>	3.480	3.000	0.996	2	6
<i>Facilitating Conditions</i>	3.699	3.625	1.114	1.5	6
<i>Hedonic Motivation</i>	4.412	4.667	1.303	0	6
<i>Habit</i>	2.559	2.667	1.474	0	6

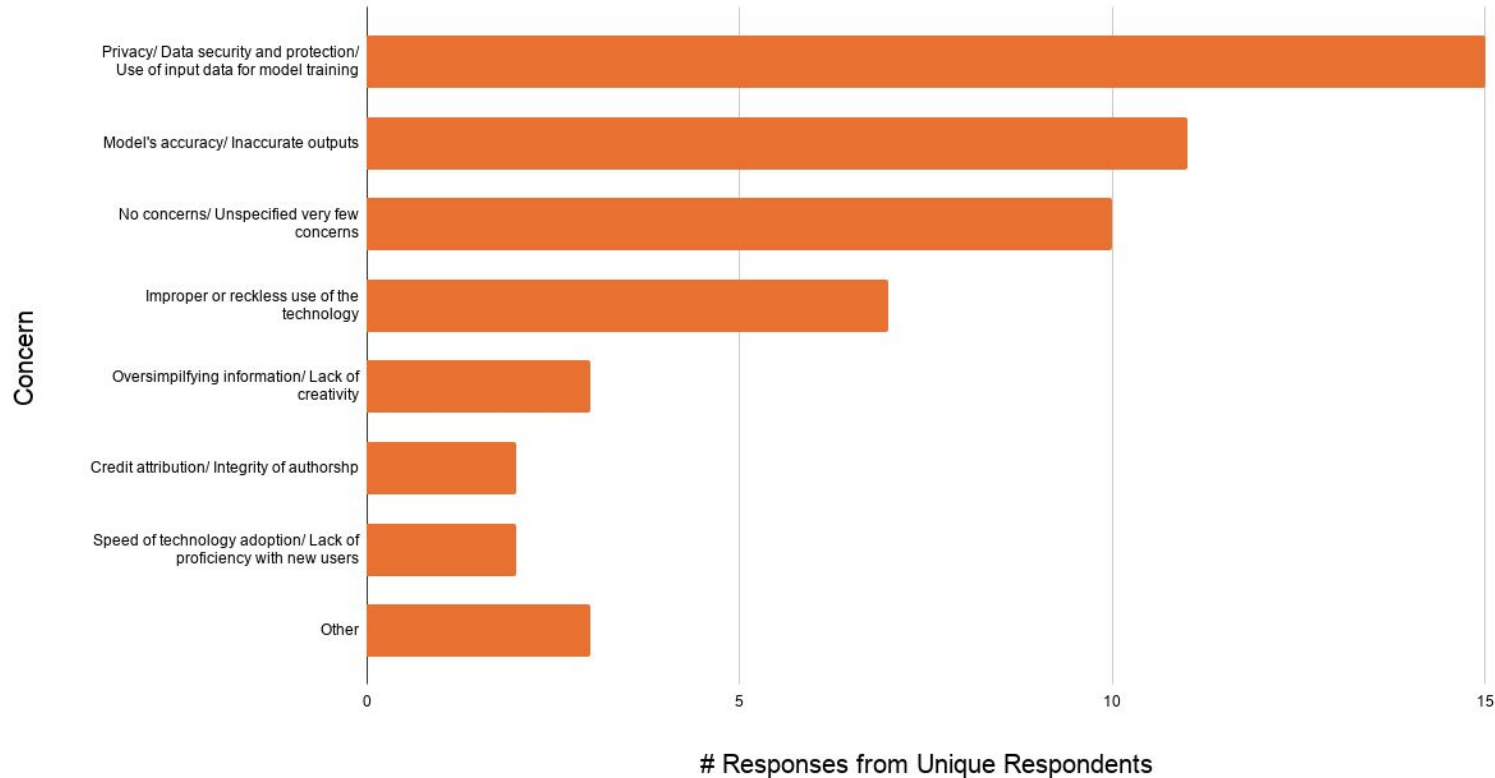
Content Analysis

How do you currently use applications that incorporate GenAI in your day-to-day work?



Content Analysis

What concerns or worries do you have in your day-to-day work regarding the use of applications that incorporate GenAI?



Proposed Design Features

Proposed Features to Promote GenAI Literacy

- Help identifying AI's strengths and weaknesses (Long & Magerko, 2020)
 - Side bars featuring information about the "skill level" of the AI
 - Most models now have a thinking feature which provides some feedback about what the model did to respond to the prompt
 - Adding a mental model which allows the model to flag its actions with a confidence level
- Bias warnings (Ethics) (Long & Magerko, 2020)
 - For data and tasks where there are known biases (i.e. job hiring)
 - The model can flag a task as sensitive
- Usage Tracker
 - A simple timer of how long the user spends actively using the page
 - What tasks is the user frequently performing
 - The ability to rate model performance on a given task and refer back to that metric

Proposed Features to Promote GenAI Literacy

- Opportunities to teach the AI (Long & Magerko, 2020)
 - Choose a predefined task at the start of a conversation and receive prompting help
 - Have a phased conversation
 - Choose a task category
 - Upload relevant source materials
 - Set memory requirements isolated to that task
 - Begin prompting with support
 - Evaluate output
 - Iterate
 - Within each step have the opportunity to provide structured prompts that help the user obtain the desired output from the GenAI

Conclusion

Key Takeaways

- Participants do not have complete GenAI literacy
 - They do not have a complete understanding of what GenAI can and cannot do
 - Their confidence in the output of the models is misplaced, and they could potentially exhibit an improper amount of trust in GenAI
- High and consistent measures of acceptance constructs **EXCEPT HABIT**
 - Participants do not think they have a habit despite the overwhelming majority reporting daily or weekly use
- Their chosen tasks leave openings for critical errors to be made
- Their concerns demonstrate some understanding of GenAI, but a troubling number of respondents state they have no concerns at all
 - Could be a demonstration of lack of caution
- GenAI applications could benefit from a level of structure that encourages caution and a correct amount of trust being put into the system

Thanks for your attention!

Questions?

References

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